

IoT Network Hybrid Cryptography Standards

1. Algorithm Selection: Resource-constrained IoT devices must utilize ASCON for lightweight authenticated encryption.
2. Key Exchange: Elliptic Curve Cryptography (ECC) shall be used for secure key exchange protocols.
3. Prohibited Architectures: A pure ECC model is strictly prohibited for all communication phases due to excessive computational overhead.
4. Performance Goals: The hybrid ASCON-ECC framework aims to reduce device battery consumption by 30% compared to legacy AES deployments.